

Ex: Synthetic Data Easy QFT

Example

Consider a complex scalar field Φ in four spacetime dimensions described by the Lagrangian:

$$\mathcal{L} = (\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) - V(\Phi) \quad (15)$$

with the potential

$$V(\Phi) = \lambda \left(|\Phi|^2 - \frac{v^2}{2} \right)^2 \quad (16)$$

where $\lambda > 0$ and $v > 0$ are real parameters. This theory has a global $U(1)$ symmetry, $\Phi \rightarrow e^{i\alpha} \Phi$. The potential induces spontaneous symmetry breaking (SSB). We choose the vacuum expectation value (VEV) to be real: $\langle \Phi \rangle = \frac{v}{\sqrt{2}}$. The quantum fluctuations around this VEV are parameterized by real scalar fields $h(x)$ (the massive mode) and $\pi(x)$ (the massless Goldstone boson) as:

$$\Phi(x) = \frac{1}{\sqrt{2}}(v + h(x) + i\pi(x)) \quad (17)$$

The conserved Noether current associated with the $U(1)$ symmetry is given by:

$$J^\mu(x) = i \left((\partial^\mu \Phi^\dagger) \Phi - \Phi^\dagger (\partial^\mu \Phi) \right) \quad (18)$$

According to Goldstone's theorem, the massless mode $\pi(x)$ is created from the vacuum by the action of this current. The matrix element of the current between the vacuum $|0\rangle$ and a one-Goldstone boson state $|\pi(p)\rangle$ with momentum p is parameterized by a constant F , known as the decay constant:

$$\langle 0 | J^\mu(x) | \pi(p) \rangle = -i F p^\mu e^{-ip \cdot x} \quad (19)$$

Task: By expanding the current $J^\mu(x)$ in terms of the fields $h(x)$ and $\pi(x)$, determine the decay constant F . Find the numerical coefficient C in the relation $F = C \cdot v$.

Conventions: The one-particle states $|\pi(p)\rangle$ are normalized such that the matrix element of the field operator is $\langle 0 | \pi(x) | \pi(p) \rangle = e^{-ip \cdot x}$.

Goal

Simple tasks for small models

Features

- Ample background context
- Simple Tasks
- One task per problem
- Few Calculations

Dataset Sizes

- All synthetic datasets have 80 problem validation sets (20 per domain difficulty)
- 4602 synthetic problems generate
- All generation cost ~\$2500

Filtering Stage	Fully Synthetic Pipeline			Human-Adapted Pipeline	
	Easy	Medium	Hard	Pedagogy	arXiv
Initial Count (Gen./Seeds)	1,452	1,650	1,500	564	397
Passed QC	1,350	1,441	908	480	333
Passed QC + GPT-5	1,106	1,092	631	NA	NA